

Topic 2: HTML — The Skeleton of the Web

1 Real-World Analogy

HTML is like architectural structure:

Building	HTML
Walls	<div>
Doors	<a>
Labels	<label>
Blueprint	DOM

Basic Structure

```
<!DOCTYPE html>
```

```
<html>
```

```
  <head>
```

```
    <title>Page</title>
```

```
  </head>
```

```
  <body>
```

```
    <h1>Hello</h1>
```

```
  </body>
```

```
</html>
```

Important Tags to Master

Tag	Purpose
section	Logical grouping
article	Independent content
nav	Navigation
form	User input
button	Actions

Interview Questions

? What is the DOM?

The DOM is a tree-structured, object-based representation of an HTML document created by the browser, which allows programming languages like JavaScript to dynamically access and manipulate the content, structure, and styles of a webpage.

? Difference Between id and class?

Id is a unique identifier used for one element, while class is a reusable label used to group multiple elements for styling or behaviour. Classes are preferred for scalability; IDs are reserved for unique cases.

Q. Explain why this is bad:

```
<div><div><div></div></div></div>
```

Because it creates **unnecessary nesting with no meaning**, making the DOM heavier, harder to maintain, and worse for performance and accessibility.

Q. why semantics are Important

1 Better Accessibility

Screen readers understand structure and help visually impaired users navigate pages correctly.

2 Improved SEO

Search engines understand your content hierarchy → better ranking.

3 Cleaner, Readable Code

Developers instantly know what each part does (no “div soup”).

4 Easier Maintenance

Structured code is easier to debug, scale, and collaborate on.

5 Built-in Browser Meaning

Browsers can interpret roles automatically (no extra work needed).

Q. What Is aria-label and Why It's Used

ARIA (Accessible Rich Internet Applications) attributes help when HTML alone is **not enough** to describe something to assistive technologies.

aria-label gives an element a **spoken description** for screen readers.

e.g.

```
<button aria-label="Close Menu">X</button>
```